

Single-stage Sounding Rocket

A Sounding rocket is a type of rocket to carry out simple missions in sub-orbital trajectory. Most sounding rockets are single-stage but some are multi-stage. The Payload is usually housed in the nose cone, which separates during the descent.

$$M_p = (M_f + M_o)$$

$$\text{Total Impulse (I)} = \int T dt = \underline{M_p U_{eq}}$$

where M_p is total mass of Propellant
[Both fuel & oxidizer]

$$\therefore M_p = \int m dt$$

Assumptions :-

- (1) vertical flight $\theta = 90^\circ$
- (2) Doesn't reach orbit
- (3) No drag
- (4) constant U_{eq} & g_e

$$F = Ma$$

$$Ma = T - \cancel{D} - mg_e \sin \theta$$

$$a = \frac{T}{M} - g_e \sin \theta$$

$$\left(\frac{du}{dt} = \left(\frac{T}{M} - g_e \sin 90^\circ \right) \right) dt$$

$$\cancel{du = -U_{eq} \frac{dM}{M} - g_e dt}$$

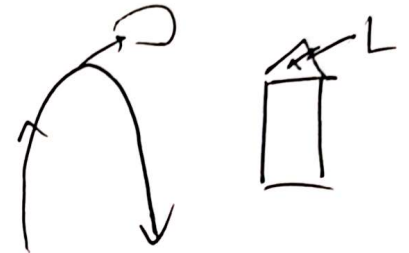
$$\frac{T}{M} = \frac{\dot{m} \left(U_{eq} + \frac{dU_{eq}}{dt} \right)}{M}$$

$$= \frac{U_{eq}}{M} \dot{m}$$

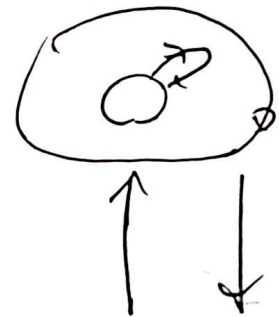
$$\frac{T}{M} = \frac{U_{eq}}{M} \left(-\frac{dM}{dt} \right)$$

$$\frac{T}{M} = \frac{\dot{m} U_{eq}}{M}$$

$$= -\frac{dM}{dt} \frac{U_{eq}}{M}$$



Suborbital flight



$$du = -\frac{u_{eq}}{M} dM - g_e dt$$

$$\frac{du}{dt} = \frac{T}{M} - g_e$$

$$\frac{du}{dt} = \frac{T}{M} - g_e$$

$$\frac{du}{dt} = \frac{u_{eq}}{M} \frac{dM}{dt} - g_e$$

Integrating over time (t_b), Propellant exhausted

$$u(t) = -u_{eq} \ln\left(\frac{M(t)}{M_0}\right) - g_e t$$

$$M(t) = M_0 - \dot{m} t$$

$$= M_0 - \frac{M_0 - M_b}{t_b} t$$

$$\begin{cases} \dot{m} = \frac{M_0 - M_b}{t_b} \text{ as} \\ \dot{m} t_b = M_0 - M_b = M_p \end{cases}$$

$$u(t) = -u_{eq} \ln\left[\frac{M_0 - \left(\frac{M_0 - M_b}{t_b}\right)t}{M_0}\right] - g_e t$$

$$u(t) = -u_{eq} \ln\left[1 - \left(1 - \frac{M_b}{M_0}\right) \frac{t}{t_b}\right] - g_e t$$

$$\rightarrow \left(\because \frac{M_b}{M_0} = \frac{1}{R}\right)$$

$$R = \frac{M_0}{M_b}$$

$$\frac{1}{R} = \frac{M_b}{M_0}$$

$$u(t) = -u_{eq} \ln\left[1 - \left(1 - \frac{1}{R}\right) \frac{t}{t_b}\right] - g_e t$$

at time $t = t_b$

$$u(b) = -u_{eq} \ln\left(\frac{1}{R}\right) - g_e t_b = u_{eq} \ln R - g_e t_b$$

Velocity at burnout

Already derived

$$= u_{eq} \ln\left(\frac{1+\lambda}{\epsilon+1}\right) - g_e t_b$$

$\underbrace{\quad}_{= g_e I_{sp}} \quad \underbrace{\quad}_{R}$

h_b - Altitude attained at burnout

$$\frac{dh}{dt} = u$$

$$h_b = \int_0^{t_b} u(t) dt$$

$$= \int_0^{t_b} \left(-u_{c2} \ln \left[1 - \left(1 - \frac{1}{R} \right) \frac{t}{t_b} \right] - g_c t \right) dt$$

$$h_b = \int_0^{t_b} \left(-u_{c2} \ln(x) - g_c t \right) dt$$

we get: $\rightarrow \int_0^{t_b} -u_{c2} \frac{\ln(x) dx}{-(1-\frac{1}{R})\frac{t}{t_b}} - \int_0^{t_b} g_c t dt$

$$= \frac{u_{c2} t_b R}{R-1} \int_0^{t_b} \ln(x) dx - g_c \frac{t_b^2}{2}$$

$$= \frac{u_{c2} t_b R}{R-1} \left[\underline{x \ln x - x} \right]_0^{t_b} - \frac{g_c t_b^2}{2}$$

$$\text{Let } x = 1 - \left(1 - \frac{1}{R} \right) \frac{t}{t_b}$$

$$dx = - \left(1 - \frac{1}{R} \right) \frac{1}{t_b} dt$$

$$\frac{dx}{-(1-\frac{1}{R})\frac{1}{t_b}} = dt$$

$$t=0 \rightarrow x=1$$

$$t=t_b \rightarrow x = \frac{1}{R}$$

$$\int \ln(x) dx = x \ln x - x$$

$$= \frac{u_{c2} t_b R}{R-1} \left[\frac{\left(1 - \left(1 - \frac{1}{R} \right) \frac{t}{t_b} \right)}{x} \ln \left(\frac{1 - \left(1 - \frac{1}{R} \right) \frac{t}{t_b}}{x} \right) - \frac{1 + \left(1 - \frac{1}{R} \right) \frac{t}{t_b}}{x} \right]_0^{t_b} - \frac{g_c t_b^2}{2}$$

$$= \frac{u_{c2} t_b R}{R-1} \left[\ln \left(1 - \left(1 - \frac{1}{R} \right) \frac{t}{t_b} \right) - \left(1 - \frac{1}{R} \right) \frac{t}{t_b} \ln \left(1 - \left(1 - \frac{1}{R} \right) \frac{t}{t_b} \right) - 1 + \left(1 - \frac{1}{R} \right) \frac{t}{t_b} \right]_0^{t_b} - \frac{g_c t_b^2}{2}$$

$$= \frac{u_{c2} t_b R}{R-1} \left[\ln(x - x + \frac{1}{R}) - \left(1 - \frac{1}{R} \right) \ln \left(\frac{1}{R} \right) - \cancel{x} + \cancel{x} - \frac{1}{R} - (-1) \right] - g_c \frac{t_b^2}{2}$$

F
↓
a
↓
u
↓
h
↓
h_{max}

so far
we write T. Eq

$$= \frac{u_{e2} t_b R}{R-1} \left[\cancel{\ln\left(\frac{1}{R}\right)} - \cancel{\ln\left(\frac{1}{R}\right)} + \frac{1}{R} \ln\left(\frac{1}{R}\right) - \frac{1}{R} + 1 \right] - g_c \frac{t_b^2}{2}$$

obtain $a = \frac{du}{dt}$

$$= \frac{u_{e2} t_b R}{R-1} \left(-\frac{\ln R}{R} + \frac{R-1}{R} \right) - g_c \frac{t_b^2}{2}$$

$$\int \frac{du}{dt} \rightarrow u$$

$$\int u dt \rightarrow h$$

$$h_b = -u_{e2} t_b \frac{\ln R}{R-1} + u_{e2} t_b - g_c \frac{t_b^2}{2} \rightarrow \text{this is only valid at burnout} \quad (*)$$

this is a special case of (*) putting $t = t_b$

for $0 < t < t_b$

$$h = \frac{u_{e2} t_b R}{R-1} \left[\ln\left(1 - \left(1 - \frac{1}{R}\right) \frac{t}{t_b}\right) - \left(1 - \frac{1}{R}\right) \frac{t}{t_b} \ln\left(1 - \left(1 - \frac{1}{R}\right) \frac{t}{t_b}\right) - 1 + \left(1 - \frac{1}{R}\right) \frac{t}{t_b} \right] - \left(\frac{g_c t^2}{2} \right)_0^t$$

$$h(t) = \frac{u_{e2} t_b R}{R-1} \left[\left(1 - \left(1 - \frac{1}{R}\right) \frac{t}{t_b}\right) \ln\left(1 - \left(1 - \frac{1}{R}\right) \frac{t}{t_b}\right) - \left(1 - \left(1 - \frac{1}{R}\right) \frac{t}{t_b}\right) + 1 \right] - \frac{1}{2} g_c t^2 \quad (*)$$

for $h_{max} \Rightarrow$

$$k.E = \frac{1}{2} M V^2$$

$$k.E = \frac{1}{2} M_b V_b^2$$

$$K.E = P.t$$

$$\frac{1}{2} M_b u_b^2 = \overbrace{M_b g_c (h_{max} - h_b)}^{W_b}$$

$$h_{max} = h_b + \frac{1}{2} \frac{u_b^2}{g_c}$$

$$E_b = E_{max} = P.E)_{max}$$

$$\frac{1}{2} M_b u_b^2 + M_b g h_b = M_b g h_{max}$$

$$u_b = u_{e2} \ln R - g_c t_b$$

$$\therefore h_{max} = -u_{e2} t_b \frac{\ln R}{R-1} + u_{e2} t_b - g_c \frac{t_b^2}{2} + \frac{u_b^2}{2g_c}$$

$$h_{max} = \underbrace{-u_{e2} t_b \frac{\ln R}{R-1} + u_{e2} t_b - g_c \frac{t_b^2}{2}}_A + \frac{(u_{e2} \ln R)^2 - 2u_{e2} \ln R g_c t_b + g_c^2 t_b^2}{2g_c}$$

$$= -u_{e2} t_b \frac{\ln R}{R-1} + u_{e2} t_b - \frac{g_c t_b^2}{2} + \left(\frac{u_{e2}^2 (\ln R)^2}{2g_c} - u_{e2} \ln R t_b + \frac{g_c t_b^2}{2} \right)$$

A

$$= -u_{e2} t_b \left(\frac{\ln R}{R-1} - 1 + \ln R \right) + \frac{u_{e2}^2 (\ln R)^2}{2g_c}$$

$$= -u_{e2} t_b \left(\frac{\ln R - R + 1 + R \ln R - \ln R}{(R-1)} \right) + \frac{u_{e2}^2 (\ln R)^2}{2g_c}$$

$$= -u_{e2} t_b \left(\frac{R \ln R}{R-1} - \frac{(R-1)}{R-1} \right) + \frac{u_{e2}^2 (\ln R)^2}{2g_c}$$

$$h_{\max} = -u_{e2} t_b \left(\frac{R \ln R}{R-1} - 1 \right) + \frac{u_{e2}^2 (\ln R)^2}{2g_c}$$

$h_{\max}$ is function of:

$$\left\{ \begin{array}{l} u_{e2} \\ t_b \\ R \\ g_c \end{array} \right.$$

(B) $u(t) = -u_{e2} \ln \left[1 - \left(1 - \frac{1}{R} \right) \frac{t}{t_b} \right] - g_c t$

From 2 pages
age

$$a(t) = \frac{du(t)}{dt} = \frac{-u_{e2} \left[-\left(1 - \frac{1}{R} \right)^{1/t_b} \right]}{1 - \left(1 - \frac{1}{R} \right) \frac{t}{t_b}} - g_c =$$

for $t \leq t_b$

diff from (B)

Chemical Rocket Performance Single-stage

(Check Previous PDF for more info)

$$M_0 = M_p + M_b = M_p + M_L + M_S$$

$$R = \frac{M_0}{M_b} = \frac{M_0}{M_S + M_L}$$

$$\lambda = \frac{M_L}{M_0 - M_L} = \frac{M_L}{M_p + M_S}$$

$$\varepsilon = \frac{M_S}{M_p + M_S} = \frac{M_b - M_L}{M_0 - M_L} = \frac{M_b - M_L}{M_0 - M_L}$$

$$\ln R = \ln \left(\frac{1+\lambda}{\varepsilon+\lambda} \right) = \ln \left(\frac{M_0}{M_b} \right) = \ln \left(\frac{M_0}{M_0 - M_p} \right) = \frac{\Delta u}{u_{ez}}$$

$$-\ln \left(\frac{M_0 - M_p}{M_0} \right) = \frac{\Delta u}{u_{ez}}$$

$$\ln \left(1 - \frac{M_p}{M_0} \right) = -\frac{\Delta u}{u_{ez}}$$

$$\Rightarrow \frac{M_p}{M_0} = 1 - e^{-\frac{\Delta u}{u_{ez}}}$$

$$M_S = M_{eng} + M_{tank}$$

$$M_L = M_0 - \underbrace{(M_{eng} + M_{tank})}_{M_S} - M_p$$

$$\frac{M_L}{M_0} = 1 - \frac{M_{eng}}{M_0} - \frac{M_{tank}}{M_0} - \frac{M_p}{M_0}$$

$$= 1 - \frac{M_{eng}}{M_0} - \frac{M_{tank} \cdot \frac{M_p}{M_0}}{\frac{M_p}{M_0}} - \frac{M_p}{M_0}$$

$$= 1 - \frac{M_{eng}}{M_0} - \frac{M_p}{M_0} \left(1 + \frac{M_{tank}}{M_p} \right)$$

$$\frac{M_{L_{tank}}}{M_0} = \frac{M_{tank}}{M_p} \frac{M_p}{M_0}$$

$$\frac{M_L}{M_0} = 1 - \frac{M_{eng}}{M_0} - \left(1 + \frac{M_{tank}}{M_P}\right) \left(1 - e^{-\frac{\lambda M_L}{u_{ez}}}\right)$$

$$\frac{M_P}{M_0} = \frac{M_{tank}}{M_0}$$

$$\varepsilon = \frac{M_S}{M_P + M_S} = \frac{M_{eng} + M_{tank}}{M_P + M_{eng} + M_{tank}} = \frac{\frac{M_{eng}}{M_0} + \frac{M_{tank}}{M_P} \left(\frac{M_P}{M_0}\right)}{\frac{M_P}{M_0} + \frac{M_{eng}}{M_0} + \frac{M_{tank}}{M_0}}$$

$$\varepsilon = \frac{\frac{M_{eng}}{M_0} + \frac{M_{tank}}{M_P} \left(1 - e^{-\frac{\lambda M_L}{u_{ez}}}\right)}{\left(1 - e^{-\frac{\lambda M_L}{u_{ez}}}\right) \left(1 + \frac{M_{tank}}{M_P}\right) + \frac{M_{eng}}{M_0}} = \frac{M_P}{M_0}$$

Example pg: 482

$$u_{ez} = 3048 \text{ m/s}$$

$$M_L = 1000 \text{ kg}$$

$$M_0 = 15000 \text{ kg}$$

$$M_S = 2000 \text{ kg}$$

$$\frac{M_{tank}}{M_0} = \frac{M_{tank}}{M_P} \left(\frac{M_P}{M_0}\right) = \frac{M_{tank}}{M_P} \left(1 - e^{-\frac{\lambda M_L}{u_{ez}}}\right)$$

$$\varepsilon = ?$$

$$\lambda = ?$$

$$\Delta u = ?$$

$$M_P = M_0 - M_L - M_S = 15000 - 1000 - 2000 = 12000 \text{ kg}$$

$$\varepsilon = \frac{M_S}{M_P + M_S} = \frac{2000}{12000 + 2000} = 0.1428$$

$$\lambda = \frac{M_L}{M_0 - M_L} = \frac{1000}{15000 - 1000} = 0.0714$$

$$\Delta u = u_{ez} \ln R = 3048 \ln \left(\frac{1 + \lambda}{1 + \varepsilon} \right) = 3048 \ln \left(\frac{1 + 0.0714}{1 + 0.1428} \right)$$

$$= 4906.76 \text{ m/s}$$

$$M_{S1} = 2000 - 410.42 = 1589.58$$

$$\varepsilon = 0.1429$$

$$\Rightarrow R = \frac{1+\lambda}{\lambda+\varepsilon} = 2.745$$

$$u_n = u_{eq} \ln \left(\prod_{i=1}^n R_i \right) = u_{eq} \sum_{i=1}^n \ln R_i$$

$$= u_{eq} \ln (2.745 \times 2.745) = 2 u_{eq} \ln (2.745)$$

$$\therefore u = 6156.55 \text{ m/s}$$

Note:-

$$\frac{u_n}{u_{eq}} = n \ln \left(\frac{1+\lambda}{\varepsilon+\lambda} \right) \rightarrow 10.34$$

$$\frac{M_{01}}{M_L} = \left(\frac{1+\lambda}{\lambda} \right)^n$$

from 10.36

$$\frac{1+\lambda}{\lambda} = \left(\frac{M_{01}}{M_L} \right)^{1/n} \quad (H)$$

$$\Rightarrow \frac{1}{\lambda} = \left(\frac{M_{01}}{M_L} \right)^{1/n} - 1 \quad (G)$$

Similarly stage rocket

$$\varepsilon \quad \lambda \quad I_{sp}$$

$$\frac{M_{01}}{M_L} = \prod_{i=1}^n \frac{1+\lambda_i}{\lambda_i}$$

10.34

$$\frac{u_n}{u_{eq}} = n \ln \frac{1+\lambda}{\varepsilon+\lambda}$$

from 10.34

$$\frac{u_h}{u_e} = n \ln \left\{ \frac{\frac{1+\lambda}{\lambda}}{\frac{\varepsilon+\lambda}{\lambda}} \right\}$$

from (H)

$$= n \ln \left\{ \frac{\left(\frac{M_{01}}{M_L}\right)^{\frac{1}{n}}}{\varepsilon\left(\frac{1}{\lambda}\right) + 1} \right\}$$

$$= n \ln \left\{ \frac{\left(\frac{M_{01}}{M_L}\right)^{\frac{1}{n}}}{\varepsilon \left[\left(\frac{M_{01}}{M_L}\right)^{\frac{1}{n}} - 1 \right] + 1} \right\}$$

(H)

(G)

$$\frac{u_n}{u_e} = n \ln \left\{ \frac{\left(\frac{M_{01}}{M_L}\right)^{1/n}}{\varepsilon \left[\left(\frac{M_{01}}{M_L}\right)^{1/n} - 1\right] + 1} \right\}$$

without gravitational force

if we consider gravitational force then

$$\frac{u_n + g e t_b \cos \theta}{u_e} = n \ln \left\{ \frac{\left(\frac{M_{01}}{M_L}\right)^{1/n}}{\varepsilon \left[\left(\frac{M_{01}}{M_L}\right)^{1/n} - 1\right] + 1} \right\}$$

if $n \rightarrow \infty$

limit $\lim_{n \rightarrow \infty} u_n = u_e$ $\lim_{n \rightarrow \infty} \left\{ -n \ln \left(\frac{\varepsilon \left(\frac{M_{01}}{M_L}\right)^{1/n} - \varepsilon + 1}{\left(\frac{M_{01}}{M_L}\right)^{1/n}} \right) \right\}$

limit $\lim_{n \rightarrow \infty} u_n = u_e$ $\lim_{n \rightarrow \infty} \left\{ -n \ln \left[\varepsilon + \frac{1 - \varepsilon}{\left(\frac{M_{01}}{M_L}\right)^{1/n}} \right] \right\}$

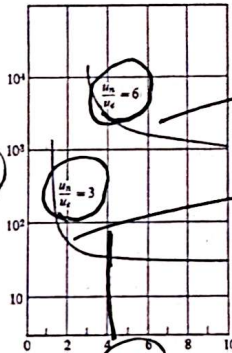
$$\propto \left[\frac{1}{\infty} \right]$$

Verify

$$u_n \approx u_e (1 - \varepsilon) \ln \left(\frac{M_{01}}{M_L} \right)$$

for $\varepsilon = 0.10$

$$\frac{M_{01}}{M_L}$$



$$\frac{u_n}{u_e} = 6$$

$$\frac{u_n}{u_e} = 3$$

the lower $\frac{M_{01}}{M_L}$ is better

Example:

$$\text{if } \frac{M_{01}}{M_{\infty}} = 15, \quad \underline{\varepsilon = 0.143} ; \quad \underline{u_e = 3048 \text{ m/s}}$$

$$\lim_{n \rightarrow \infty} \underline{u_n = ?}$$

$$= u_e (1 - \varepsilon) \ln \left(\frac{M_{01}}{M_{\infty}} \right)$$

$$= 3048 (1 - 0.143) \ln(15)$$

$$= \boxed{7073.79 \text{ m/s}}$$